PPP-P-420B
August 15, 1972
SUPERSEDING
Fed. Spec. PPP-P-420A
March 23, 1959

FEDERAL SPECIFICATION

PLUGS AND FLANGES
(FOR DRUM CLOSURES)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers new flanges and new plugs. For intended use see paragraph 6.1.

1.2 Types and classes. Flanges and plugs shall be furnished in the following types and classes, as specified (see 6.2).

Flanges:

Type I - (Octagon base).

Classes:
A - 2 inch - 11-1/2 threads (internal).
B - 3/4 inch - 14 threads (internal).

Type II - (Serrated base).

Classes:
A - 2 inch - 11-1/2 threads (internal).
B - 3/4 inch - 14 threads (internal).

Type III - (Crimp-on plastic).

Classes:
A - 2 inch - 11-1/2 threads (internal).
B - 3/4 inch - 14 threads (internal).

Plugs:

Type I - (Round head).

Classes:
C - 2 inch - 11-1/2 threads (external).
D - 3/4 inch - 14 threads (external).

Type II - (Hexagon head).

Classes:
C - 2 inch - 11-1/2 threads (external).
D - 3/4 inch - 14 threads (external).

Type III - (External).

Classes:
C - 2 inch - 11-1/2 threads (external).
D - 3/4 inch - 14 threads (external).
E - press in

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

QQ-Z-325 - Zinc Coating. Electrodeposited Requirements For.
PPP-B-591 - Boxes, Fiberboard, Wood Cleated.
PPP-B-636 - Boxes, Fiberboard.
PPP-D-723 - Drums, Fiber.
PPP-D-729 - Drums: Metal 55-gallon (For shipment of noncorrosive liquids).

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-G-432 - Gaskets, Synthetic Rubber (For Fuel and Lubricant Containers and Accessories).

Military Standards:
MIL-STD-105 - Sampling Procedures and Tables of Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

Laws and Regulations:
49CFR 171 to 191
Transportation

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

American National Standards Institute

ANSI B 2.1 - 1968

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Type I and type II flanges and plugs and type III crimping rings. Flanges, plugs and crimping rings shall be made of low carbon steel. Steel strip thickness dimensions shall be the nominal; individual strips shall be subject to the "Manufacturers' Standard" of the American Iron and Steel Institute covering variations and tolerances.

3.1.2 Type III flanges and plugs. Flanges and plugs shall be molded of high quality commercial polyethylene or comparable material.

3.1.3 Gasket material. The gasket for type I and II, plugs must be made of material that complies with MIL-G-432. All type I and II plugs must have gaskets applied to them before they are used as a closure in a drum.

3.2 Type I and II flanges. Flanges shall be a mechanical inserted type approved by the Bureau of Explosives and shall make a tight fit when inserted (4.3.2.1). If the 2-inch flange projects inward beyond the inner surface of the drum, it shall have two holes, not more than 5/16 nor less than 1/8 inch in diameter, to permit drainage thr-cugh the projection.

3.2.1 Threads. All flanges shall have not less than three complete threads of modified American National Standard Institute pipe threads in accordance with table I.

TABLE I Flange thread sizes before insertion

Class	Size	Number of threads per inch	Major diameter		Pitch diameter		Minor diameter	
			Max.	Min.	Max.	Min.	Max.	Min.
	Inches		Inches	Inches	Inches	Inches	Inches	Inches
A	2	11-1/2	----	2.3495	2.3150	2.2930	2.2628	2.2428
B	3/4	14	----	1.0324	1.0045	0.9860	0.9648	0.9448

3.2.1.1 Class A of type I and II. Before insertion in the drum head, class A flanges shall have 2-inch 11-1/2 threads (internal) per inch in accordance with paragraph 3.2.1.

3.2.1.2 Class B of type I and II. Before insertion in the drum head, class B flanges shall have 3/4-inch 14 threads (internal) per inch in accordance with paragraph 3.2.1.

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3.2.2 Zinc plating. The finished flanges shall be electro-zinc plated in accordance with type II (color may be clear chromats) of QQ-Z-325 except that the zinc plating thickness shall be 0.0003 inch minimum on flat surfaces. When specified (see 6.2), plugs and flanges shall not be zinc plated on any portion that may come in contact with military lubricating oils, hydraulic fluids or fuels.

3.2.3 Type I, classes A and B. Before insertion in the drum head, the flange shall have an octagonal base with an upstanding neck. The upper portion of the neck shall be formed for curling over the drum stock neck. The thickness of the metal in back of the flange threads shall be a minimum of 0.040 inch when measured from any point on the major diameter of the internal thread to the outside wall of the threaded upstanding neck of the flange. Alternatively, steel strip having a thickness of no less than 0.093 inch (see 3.1) shall be used to form the flange.

3.2.3.1 Class A. The dimension across the flats of the octagon shall be 3.068 ± .006 inches and the height of the upstanding neck above the base shall be 0.667 ± .015 inch.

3.2.3.2 Class B. The dimensions across the flats of the octagon shall be 1.720 ± .010 inches and the height of the upstanding neck above the base shall be 0.532 ± .015 inch.

3.2.4 Type II, classes A and B. Before insertion in the drum head, the flanges shall be flared or cone shape and designed so that they will flatten and grip without fracturing when installed in the drum head stock. The flanges shall be made from at least 0.105-inch strip steel (see 3.1.1) and have a flowed-on composition gasket fully cured and adhered to the top surface of the flange next to the serrations.

3.2.4.1 Class A. The outside perimeter shall have no less than 20 serrations on an O.D. of 3.170 ± .020 inches and a width of 0.188 ± .020 inch. The overall height of the flange shall be no less than 0.625 ± .020 inch.

3.2.4.2 Class B. The outside perimeter shall have no less than 16 serrations on an O.D. of 1.740 ± .020 inches and a width of 0.156 ± .020 inch. The overall height of the flange shall be 0.531 ± .020 inch.

3.2.5 Type III, class A and B. The flange shall be fastened to the preformed opening by means of a retaining ring made of 24 gauge cold rolled steel, zinc plated (see 3.2.2), or a comparable material with the same preformance characteristics. The retaining ring shall be crimped to the preformed opening by means of crimping tool.

3.2.5.1 Class A. The flange shall have a 2 inch opening with 11-1/2 threads (internal) per inch to receive a plug. The flange shall have an integral molded diaphram with a pull tab for lifting out and shall fit a preformed opening in the container stock. Opening shall be maximum 2-15/16 ± 1/32 inches in size with the metal curved outward.

3.2.5.2 Class B. The flange shall have a 3/4 inch 14 internal threads per inch opening to receive a plug. The flange shall have an integral molded diaphram at the base of the 3/4 inch opening which must be removed for dispensing. The preformed opening in the container stock shall be 2-3/8 inches ± 1/32.

3.3 Plugs.

3.3.1 Type I and II plug construction. Plugs shall be designed so that they may be inserted or removed by means of a simple tool. The 2-inch plug shall have a wrenching insert projection welded to the bottom of the sump of the plug (minimum 4 corners) or have a wrenching device formed as part of the plug. The 0.750-inch plug shall be hex-headed or have a wrenching insert projection welded to the bottom of the sump of the plug (minimum 2 spots) or have a wrenching device formed as part of the plug.

3.3.1.1 Threads. The overall height of the plug shall allow for three (3) or more complete threads to be engaged when the plug with gasket in place, is screwed tightly (see 4.3.2.1 and 6.1) into an inserted flange. Threads shall be modified American National Standard in accordance with Table II.

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TABLE II. Plug thread sizes

Class	Size	Number of threads per inch	Major diameter		Pitch diameter		Minor diameter	
			Max.	Min.	Max.	Min.	Max.	Min.
	Inches		Inches	Inches	Inches	Inches	Inches	Inches
C	2	11-1/2	2.3395	2.3195	2.2830	2.2610	2.2328	----
D	3/4	14	1.0274	1.0074	0.9810	0.9630	0.9308	----

3.3.1.2 Zinc plating. The finished plugs shall be zinc plated in accordance with paragraph 3.2.2.

3.3.2 Type I, class C. The 2-inch plug shall be made from a 0.062 inch thick steel (see 3.1.1) stamping, have a round head diameter of 2.400 ± .030 and a recessed gasket seat immediately beneath with a diameter of 2.215 ± .015 inches and a width from .090 to 0.125 inch.

3.3.3 Type I, class D. The 3/4 inch plug shall be made from a 0.062 inch thick steel (see 3.1) stamping, have a round head diameter of 1.120 ± .020 and a recessed gasket seat immediately beneath with a diameter of .895 ± .015 inches and a width from .090 to .125 inch.

3.3.4 Type II, class C. The 2-inch plug shall have a hexagon shaped head with a dimension of no less than 2.875 inches across the flats and these flats shall have round corners. The plug shall be made from no less than 0.093 nor more than 0.105 inch thick strip steel (3.1) and have no less than two sealing grooves pressed into the gasket seat or underside of the flats with a depth no less than 0.011 inch. The diameter of the plug above the threaded portion shall be 15/16 ± 1/64.

3.3.5 Type II, class D. The 3/4 inch plug shall have a hexagon shaped head with a dimension of no less than 31/64 inches across the flats and these flats shall have rounded corners. The plug shall be made from 0.093 inch strip steel (see 3.1) and have no less than two sealing grooves pressed into the gasket seat or underside of the flats with the depth of the grooves no less than 0.011 inch. The diameter of the plug above the threaded portion shall be 1.0 ± 1/64 inch.

3.3.6 Type III, class C. The plug of the plastic closure shall be made of high density polyethylene or comparable material. It shall have 2 inch 11-1/2 external threads per inch to mate with flange and shall have an opening in its center with 3/4 inch 14 internal threads per inch and an integral molded diaphram (See 3.3.7). The plug shall have four internal recesses or four external lugs for attaching a wrench or simple tool for tightening or removal.

3.3.7 Type III, class D. The 3/4 inch plug shall be made of material specified in 3.1.2. The plug shall have 3/4 inch 14 external threads per inch to mate with a type III, class B flange or type III, class C plug as required. The plug shall have a recessed area in the top to accommodate a wrench.

3.3.8 Type III, class E vent plug. The vent material shall be in accordance with paragraph 3.1.2 and shall have an integral molded in diaphram with pull tab for removing. The vent shall be designed to receive a cap by means of a snap fit and the cap shall be molded to the vent by means of a light web or strap. The vent shall fit into a prepared opening in the drum stock by means of a press fit and the hole size in the container stock shall be 0.720 inch + 0.005 or - 0.010.

3.4 DOT Specifications.

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3.4.1 Plugs and flanges inserted in drums used to ship certain specific materials which are classified under DOT regulations as dangerous or corrosive materials require special threads as specified in DOT container specifications and Federal Specification and shall supersede the requirements of this specification for these items.

3.4.2 If there is a conflict between this specification and the DOT specification for plugs and flanges ordered, the plug and flange manufacturer shall give the procurement officer a statement in writing describing the conflict and obtain instructions for resolving the conflict before proceeding with the manufacture of plugs and flanges.

3.5 Workmanship. Plugs and flanges shall be made with smooth surfaces, without sharp edges, burrs or other protrusions that will be a safety hazard to personnel handling them. Also, plugs and flanges shall meet the requirements of all subsidiary specifications as related to workmanship.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right tc perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Except where otherwise indicated, sampling for inspectance and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, the supplier is responsible for insuring that components and materials used are manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification.

4.2.2 End item inspection.

4.2.2.1 Visual examination. Examination of the end item shall be in accordance with the classification of defects set forth in table I. The lot sizes shall be expressed in units of plugs or flanges for the purpose of determining the sample size in accordance with MIL-STD-105. The sample unit for this examination shall be one completely fabricated plug or flange of the type or class specified.

TABLE I. Classification of defects in accordance with MIL-STD-105

Examine	Defects	Classification	
		Major	Minor
Finish	Not type or class specified.	X	
	Not zinc coated or area of no coating	X	
	Coated, not adherent, i.e., excessive flaking or peeling.		X
	Blisters, lumps, acid, or black spots.		X
	Rust, pitted, rough deposits, dross or flux.		X
	Tear drops or dripping on edges.		X
	Dirt, grit, or other foreign matter imbedded in finish.		X
Construction and Workmanship	Hole, cracked, fractured, or malformed.	X	
	Threads damaged, deformed or less than three threads.	X	
	Type II, class A and B flanges not flared or cone shaped.	X	
	Outside perimeter of type II, class A, flanges fabricated with less than 20 serrations.		X

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TABLE I (Con't)

Examine	Defects	Classification Major	Classification Minor
Construction and Workmanship	Outside perimeter of type II, class B flanges fabricated with less than 16 serrations.		X
	Corners of type II, classes C and D, flats not rounded.		X
	Gaskets for types I or II plugs missing.	X	
	Gaskets for types I or II plugs not forced over threads into recess.		X
	Seals for type III flanges not as specified.	X	
	Composition gasket missing on type II classes A and B.	X	

4.2.2.2 Dimensional examinations. Inspection shall be made of end item for compliance with dimensions specified. Any deviation from specified requirements shall be cause for rejection of the lot.

4.2.2.3 Inspection levels and AQL's. The acceptance quality levels (AQL's), expressed in defects per 100 units, and inspection levels (for sample size shall be as shown in table II.

TABLE II

	Inspection levels	AQL's Major	AQL's Total	AQL's Minor
For examination in 4.2.2.2 (1 class).	II	2.5	6.5	6.5
For examination in table III.	S-2	---	4.0	

4.2.2.4 Testing of the end item. Tests shall be performed in accordance with table III for the characteristics specified therein. The sample size and acceptance number applicable to each end item lot shall be indicated in table III. These characteristics are applicable to individual unit requirements.

TABLE III

Characteristic	Requirement paragraph	Test Method	Sample size	Number
Thread size.	3.2.1	4.3.1	25 each plugs and flanges	0
Resistance to salt spray.	3.3.2 3.2.2	4.3.2 4.3.3 this spec.	3 each, plugs and flanges	0
Zinc coating thickness	3.2.2 3.3.3	thickness gage	10 each, plugs and flanges.	0

4.2.3 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing, and marking as required by section 5 are complied with. Defects shall be scored in accordance with table IV below. The sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of containers in the inspection lot. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per 100 units.

TABLE IV. Classification of defects in accordance with MIL-STD-105

Examine	Defects
Markings (exterior and interior container)	Incorrect; illegible; of improper size, location sequence or method of application.
Materials	Any nonconforming component; component missing, damaged, or otherwise defective, affecting serviceability.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling, bulging, or distortion of containers.
Weight (exterior container)	Net weight exceeds requirements.

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4.3 Tests.

4.3.1 Mechanical insertion of type I or II flanges. The samples selected in 4.2.2.1 shall be inserted in a prepared drum or section of a drum using the tools and gages supplied by the flange manufacturer. A plug equipped with proper gasket (see 6.1) shall then be inserted in the assembled flange and the following torque applied:

2 inch size ---------- 30 foot-pounds.
3/4 inch size ---------- 15 foot-pounds.

The assembly shall then be checked to see that three complete threads are engaged. Any failure to insert properly, or if the flange turns or does not make a tight fit; shall be cause for rejection of the lot.

4.3.2 Mechanical insertion of type III flanges or class E vent. Any failure to insert properly, make a tight fit or leak when tested in accordance with the applicable drum specification shall be cause for rejection of the lot.

4.3.3 Thread size. The samples or leaks when tested in accordance with applicable drum specification selected in 4.2.2.1 shall be gaged with go-no-go gages or equivalent. Any failure to gage properly shall be cause for rejection of the lot.

4.3.4 Zinc plating. The plating shall pass the 36 hour salt spray test requirements specified in QQ-Z-325. Evidence of red rust on flat exterior surfaces shall constitute failure. White corrosive products of zinc as indicated in QQ-Z-325 shall not constitute failure. Slight red rust on threads shall not be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 There is no preservation or packaging requirements for these items.

5.2 Packing. Plugs and flanges shall be shipped in separate container, level A, B, or C as specified (see 6.2).

5.2.1 Level A. Plugs and flanges, of one type and size only, shall be packed in containers conforming to PPP-P-591, overseas type; PPP-B-636, type SF class weather-resistant, grade V2s; or PPP-D-723, type II, grade. Each shipping container shall be closed and strapped in accordance with the appendix of the applicable container specification. Flat steel strapping shall conform to QQ-S-781, type I or IV, class B. The net weight of contents of fiberboard boxes, cleated fiberboard boxes, and fiber drums shall not exceed 125 pounds.

5.2.2 Level B. Plugs and flanges, of one type and size only, shall be packed in containers, conforming to PPP-B-591, domestic type, style A or B; PPP-B-636, type CF, class domestic, grade 600; or PPP-D-723, type I, grade A. Each shipping container shall be closed in accordance with the appendix of the applicable container specification. The net weight of contents for fiberboard boxes, cleated fiberboard boxes, and fiber drums shall not exceed 125 pounds.

5.2.3 Level C. The plugs and flanges shall be packed in containers that will assure carrier acceptance and safe delivery at destination in compliance with the Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.3 Marking.

5.3.1 Civil agencies. In addition to special marking required by the contract or order, each shipping container shall be marked in accordance with Fed. Std. 123.

5.3.2 Military agencies. In addition to special marking required by the contract or order, each shipping container shall be marked in accordance with the requirements of MIL-STD-129.

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6. NOTES

6.1 Intended use.

6.1.1 Types I and II plugs and flanges are intended for use in Rule 40 drums of Freight Classifications and DOT Title 49 specifications for non-corrosive materials and PPP-D-729.

6.1.2 Type I plugs shall be used in type I flanges, type II plugs in type II flanges, and type III plugs in type III flanges. Plugs, gaskets and flanges shall not be interchanged.

6.1.3 Type III plugs and flanges are to be used for drums on a when specified basis.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Type and class flange and plug should be ordered to fit the corresponding unit.
(c) New gaskets should be ordered.
(d) Plug (see 1.1 and 3.3).
(e) Flange (see 1.1 and 3.2).
(f) Gasket (see MIL-G-432).
(g) Selection of application level of packing (see 5.2).
(h) Whether only one type, class, style, and size shipping container should be used for any one item on any one contract or order.
(i) When zinc plating not required (see 3.2.2).

6.3

Cross Reference

PPP-P-420A - Amend. 2	PPP-P-420B
FLANGES:	
Type I - (Octagon base) Class A Class B	Type I - (Octagon base) Classes: A B
Type II - (serrated base) Class C Class D	Type II - (Serrated base) Classes: A B
Type III - (crimp-on) Class A Class B	Type III - (crimp-on plastic) Classes: A B
PLUGS:	
Type I - (Round head) Class A Class B	Type I - (Round head) Classes: C D
Type II - (Hexagon head) Class C Class D	Type II - (Hexagon head) Classes: C D
Type III - (internal or external lug) Class A Class B	Type III - (external) Classes: C D E

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